

BANARAS HINDU UNIVERSITY

B.A. (Hons.) Semester I Examination, 2023-24

Paper No. STA-101: Statistical Methods and Probability

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) Explain the method of constructing a frequency polygon. [1]
- (b) Define the difference between nominal and ratio scale with suitable examples. [1]
-

Question 2.

- (a) Describe the characteristics to be satisfied by an ideal measure of central tendency. Show that the sum of the squares of the deviations of a set of values is minimum when taken about their mean. [1]
- (b) Describe the geometric mean of the combined groups. [1]
-

Question 3.

- (a) What do you understand by the mode of data? Derive the expression of the mode for a continuous frequency distribution. [1]
- (b) Derive the expression for the variance of the combination of two series. [1]
-

Question 4.

- (a) What do you mean by standard deviation? Show that for any discrete distribution, standard deviation is not less than the mean deviation from the mean. [1]
- (b) Describe the relative positions of mean, median and mode in the figures for positively and negatively skewed distributions. Explain the method of measuring kurtosis of a frequency distribution. [1]
-

Question 5.

- (a) What is the subjective approach to probability? Suppose E_1 and E_2 are two events and $E_2 \subseteq E_1$, then show that:
-

(i) $P(E_1 \cap E_2^c) = P(E_1) - P(E_2)$

(ii) $P(E_2) \leq P(E_1)$

[#1]

(b) State and derive the addition theorem of probability for n events.

[#1]

Question 6.

(a) What do you understand by conditional probability? State and prove Bayes' theorem.
[#1]

(b) A factory produces a certain type of product by two types of machines. Past records show that Machine I produces 40% of the items of product and Machine II produces 60% of the items. Further, 5% of the items produced by Machine I are defective, and only 1% produced by Machine II are defective. If a defective item is drawn at random, what is the probability that it was produced by Machine II?
[#1]

Question 7.

(a) Describe the multiplication theorem of probability for independent events. Suppose E_1 and E_2 are two independent events, then show that E_1^c and E_2^c are also independent.
[#1]

(b) What do you mean by a continuous distribution function? Discuss the properties of a continuous distribution function.
[#1]

Question 8.

(a) The joint probability distribution of two random variables X and Y is given below:

$Y \setminus X$	1	2	3	4	5	6
0	0	0	1/32	1/16	1/16	3/32
1	1/16	1/16	1/8	1/8	1/8	1/8
2	1/32	1/32	1/64	1/64	0	1/32

(i) Find the marginal distribution of X .

(ii) Find the conditional distribution of Y given the value of $X = 2$.

[#1]

(b) Describe the meaning of mathematical expectation of a random variable. Show that the mathematical expectation of the product of a number of independent random variables is equal to the product of their expectations.
[#1]

[Click here to visit our website](#)

[Click here to visit our contributor](#)

[Click here to visit our contributor](#)